

Project Status Report

Project Name: Detecting Disease Contacts

Team Name: Circuit Breakers

Project Manager: Robin Pottathuparambil

Team Members: Tarod Anderson, Jared Beyers, Ronil Doshi, Isaiah Maberry, Jennifer Tapia

Report Date: November 6, 2025

Reporting Period: October 10, 2025 to November 6, 2025

Management Summary

Defined milestones completed: 12 of 15 (80%)
Defined tasks completed: 0 of 0 (0%) of child tasks in work breakdown structure
Total estimated project hours used: 800 of 1600 hours (50%)
Ahead of (or Behind) schedule by: 0 labor hours, 0 days
Known defects: N/A
Staff members on project: N/A
Contingency hours remaining: 100% of 350 hours

Schedule

Initial estimated completion date: April 9, 2026
Previous estimated completion date: N/A
Current estimated completion date: April 9, 2026

Key Milestones Table

ID	Title	Planned Completion Date	Previous Forecast Completion Date	Current Forecast Completion Date	Actual Completion Date
1	Formed Team	8/21/25	8/21/25	8/21/25	8/21/25
2	Selected Project	8/28/25	N/A	8/28/25	8/28/25
3	Drafted Functional Requirements	9/12/2025	N/A	9/12/2025	10/01/2025
4	Drafted Non-Functional Requirements	9/12/2025	N/A	9/12/2025	10/01/2025
5	Status Report 1	9/11/2025	N/A	9/11/2025	9/11/2025
6	Established an initial budget	9/11/2025	N/A	9/11/2025	9/11/2025
7	Drafted Specifications	10/2/2025	N/A	10/2/2025	10/02/2025
8	Finalized Requirements	10/2/2025	N/A	10/2/2025	10/2/2025
9	Preliminary Design	10/3/2025	N/A	10/3/2025	10/03/2025
10	Finalized Specifications	10/16/2025	N/A	10/16/2025	10/16/2025

11	<i>Parts list</i>	10/9/2025	N/A	10/10/2025	11/06/2025
12	<i>Detailed Design</i>	10/17/2025	N/A	10/17/2025	10/17/2025
13	<i>Four camera assembly works with each other</i>	2/12/2026	N/A	2/12/26	
14	<i>Testing and Verification</i>	4/1/2026	N/A	4/1/2026	
15	<i>Product Complete</i>	04/9/2026	N/A	4/9/2026	

Product Size

Microcontroller: Raspberry Pi 5 + Raspberry Pi Zero 2 W

Subsystem dimensions: 7x15x2 cm

Central System dimensions: 10x20x5 cm

RAM: 8 GB

Effort

Life Cycle Activity	This Reporting Period (labor-hours)		Project to Date (labor-hours)	
	Planned Effort	Actual Effort	Planned Effort	Actual Effort
Specifications	50	100	180	160
Detailed Design	50	50	50	50
Parts Order	100	150	250	300

Cost

Life Cycle Activity	This Reporting Period		Project to Date	
	Planned Cost	Actual Cost	Planned Cost	Actual Cost
Materials	900	424.87	900	424.87
Cloud computing	100	0	100	0

Requirements Status

I D	Functional Requirements	Team Member Responsible	Effort (%)	Verification	Completed (%)
F R 1	The system should operate in both day and night conditions.	Jared Beyers	20%	Testing	5 %
F R 2	The system should trigger the camera system when it detects possible human presence.	Jared Beyers	30%	Testing	5 %

F R 3	The system should use computer vision to determine if a possible presence is a human.	Ronil Doshi	25%	Testing	10 %
F R 4	The system should algorithmically determine and record the closest distance for each contact.	Ronil Doshi	30%	Analysis	10 %
F R 5	The system should detect all humans exiting 2 meter radius.	Jared Beyers	30%	Testing	5 %
F R 6	The system should track all human subjects so long as they remain in the 2 meter radius.	Ronil Doshi	35%	Testing	10 %
F R 7	The system should record GPS location for all encounters.	Isaiah Maberry	25%	Testing	0 %
F R 8	The cloud should calculate and store data based on frequency of contacts.	Tarod Anderson	15%	Demonstration	20 %
F R 9	The system should send data to the cloud for storing and access.	Tarod Anderson	25%	Demonstration	20 %
F R 10	The smartphone app should access data from the cloud.	Tarod Anderson	25%	Demonstration	0 %
F R 11	The smartphone app should display all contacts with their respective timestamps and locations on a map, along with the total number of contacts.	Jennifer Tapia	25%	Demonstration	0 %
F R 12	The smartphone app should have a searchable list of contacts. With each contact showing data on duration, distance, timestamp, and location.	Jennifer Tapia	30%	Demonstration	10 %

F R 1 3	The smartphone app should display frequency of contacts, with filters such as per hour, per day, and per week.	Jennifer Tapia	20%	Demonstration	5 %
F R 1 4	The smartphone app shall interact with cloud database to secure user logins and protect access to contact data.	Isaiah Maberry	25%	Testing	15%
F R 1 5	The system should delete all user video after processing is complete.	Ronil Doshi	10%	Demonstration	0 %
F R 1 6	The smartphone app should display power levels of each subsystem and the central system.	Isaiah Maberry	15%	Inspection	0 %
F R 1 7	The subsystems should send data to the central system via wifi.	Isaiah Maberry	15%	Inspection	5%

ID	Non-Functional Requirements	Team Member Responsible	Effort (%)	Verification	Completed (%)
N FR 1	The system should be embedded into a wearable vest.	Tarod Anderson	25%	Inspection	0 %
N FR 2	The system should be discrete.	Tarod Anderson	10%	Inspection	0 %
N FR 3	The system should use a battery for 4-hour operation.	Isaiah Maberry	15%	Analysis	0 %
N FR 4	The system should be under 1 KG.	Ronil Doshi	10%	Inspection	0 %

N FR 5	The system shall be implemented on a PCB that ensures compact size and reliable signal routing.	Isaiah Maberry	20%	Inspection	0 %
N FR 6	All components shall be housed in enclosures to ensure protection and user comfort.	Jared Beyers	20%	Inspection	5%
N FR 7	The smartphone app should be professional, elegant, and user friendly.	Jennifer Tapia	25%	Inspection	5%

Top Five Risks

1. Possibility of parts shipment being delayed

Open Issues

- Possible complications with AWS and SQL server

Action Items

- Implement vision algorithm on Raspberry Pi
- Connect camera to Raspberry Pi
- Capture day and night photos with IR camera
- Send photos from Pi zero to Pi 5
- Calculate distance from camera and sensor data
- Algorithm for heat array sensing
- Mobile app development
- Implement methods for retrieving data from SQL server

Defects

N/A